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August 31, 2026 · 5:00 – 7:30 PM · Kai Santos

NOTES

MECH 191 — Metallurgy · Week 2 Notes · fill in blanks & key terms as you go

01 Stress & Strain

01

02

Crystal Defects

03

Grain Boundaries & Diffusion

SLIDE 2

NOTES

Portfolio Handout — Module 2

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MECH 191 — Metallurgy · Week 2 Notes · fill in blanks & key terms as you go

MECH 191 — Course & Unit Roadmap

Unit 1	Unit 2	Unit 3	Unit 4	Unit 5
Materials Fundamentals	Steel & Alloys	Testing & Quality	Manufacturing	Joining & Failure
Wk 1 Aug 24 Atomic Structure & Bonding	Wk 4 Sep 14 Carbon & Alloy Steels	Wk 7 Oct 5 Mechanical Testing	Wk 10 Oct 26 Casting	Wk 14 Nov 23 Welding Processes
Wk 2 ← here Aug 31 Mechanical Properties	Wk 5 Sep 21 Stainless & Cast Irons	Wk 8 Oct 12 NDT Methods	Wk 11 Nov 2 Forming	Wk 15 Nov 30 Weld Metallurgy & Failure
	Wk 6 Sep 28 Nonferrous & Phase Diagrams	Wk 9 Oct 19 Quality Standards	Wk 12 Nov 9 Machining	Wk 16 Dec 7 Corrosion & Wrap-up
			Wk 13 Nov 16 Powder Metallurgy	

Week 17 • FINAL EXAM

SLIDE 4 MECH 191 — Course & Unit Roadmap

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Topics

- Stress and strain — the language of mechanical behavior
- Elastic vs. plastic deformation and the yield point
- Key properties: strength, hardness, ductility, toughness, brittleness
- Crystal defects — point, line, planar, and volume
- Grain boundaries — formation, energy, and the Hall-Petch effect
- Diffusion — how atoms move through solid metal

Reference Material

Kalpakjian 6th — Ch. 2
Fundamentals of Mechanical
Behavior of Materials

Visualizers

- Stress-Strain Curve (Interactive)
- Hot vs Cold Working & Work Hardening

NOTES

Try to answer these before we dive in — we'll revisit them with answers at the end of class.

- 1 What is the difference between strength and toughness? Can a material be strong but not tough?
- 2 At the atomic level, what exactly happens at the yield point when you pull a metal specimen in tension?
- 3 Cold-working a metal (rolling, drawing) increases its strength but reduces ductility. What microstructural change explains both effects simultaneously?
- 4 A fine-grained and a coarse-grained steel have identical composition. Which yields at higher stress — and why?
- 5 A steel shaft fails well below its yield strength after thousands of rotation cycles. What failure mode is this, and what property should have been checked?

SLIDE 6 Core Questions

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When a force is applied to a metal, we need precise language to describe what happens inside it — at the grain level and at the atomic level.

$$\sigma = F / A$$

Units: Pa, MPa, ksi

$$\epsilon = \Delta L / L_0$$

Dimensionless — expressed as mm/mm, in/in, or %

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MECH 191 — Metallurgy · Week 2 Notes · fill in blanks & key terms as you go

Elastic Zone

Spring, tight bolt that can still be removed, a bent ruler that snaps back

The 'click' when a bolt is overtightened — it won't come back to correct preload

Bent paperclip, cold-formed bracket,
stamped sheet metal door panel

SLIDE 8 *Elastic vs. Plastic Deformation*

These five properties are how engineers describe and compare materials — you will encounter all of them in specifications, data sheets, and test reports.

SLIDE 9 Key Mechanical Properties

No real metal is perfect — defects exist at every scale. Point defects are disruptions at a single lattice site. Every cubic centimeter of real metal contains millions.

A missing atom — an empty site where an atom should be. The most common point defect in all metals. Number increases exponentially with temperature.

Enable atom movement (diffusion).
Essential for heat treatment, annealing,
and solid-state reactions.

An extra atom squeezed into a gap between host atoms. If the same element: self-interstitial. If a smaller foreign element: interstitial alloy.

Carbon in iron is an interstitial defect — even tiny amounts dramatically harden steel by blocking dislocation movement.

A wrong atom sitting in a lattice site, replacing a host atom. Works best when the substitute is within ~15% in size and has similar valence electrons.

Cr, Ni, Mo in steel are all substitutional — this is how stainless steel and alloy steels are made.

SLIDE 10 Crystal Defects — Point Defects

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Core question:

If dislocations make metals easier to deform, why don't we want to eliminate them? A metal with zero dislocations would need enormous stress to deform at all, then fracture suddenly with no warning — like glass.

Screw Dislocation

Burgers vector (b): perpendicular to the dislocation line

A helical distortion of the lattice — atomic planes spiral around the dislocation line like a corkscrew. Can cross-slip between slip planes, giving more paths to deform.

Burgers vector (b): parallel to the dislocation line

Like tearing paper by sliding the two halves sideways rather than pulling them apart

In practice, most dislocations are a mixture of edge and screw character — the Burgers vector is at an angle. Pure edge or pure screw are rarely found in real metals.

Burgers vector: at an angle — has both edge and screw components

Real metal dislocations wind and curve through the crystal in complex three-dimensional paths

SLIDE 11 *Crystal Defects — Line Defects (Dislocations)*

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Two-dimensional surfaces where crystal structure or orientation changes — grain boundaries are the most common and most important.

The most common planar defect. When independently nucleating crystals grow into each other during solidification, the orientation mismatch creates a disordered boundary region.

- A boundary where the crystal on each side is a perfect mirror image of the other — extremely low energy and very stable.
Two types with different origins:

- A disruption in the normal atomic stacking sequence — FCC follows ABCABC... and HCP follows ABAB...; a fault breaks this pattern. Low Stacking Fault Energy (SFE) → wide stable faults → dislocations can't cross-slip → the metal work-hardens quickly. High SFE → narrow faults → easy cross-slip → softer cold-working behavior.

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Three-dimensional imperfections large enough to detect with inspection equipment. These are the defects that cause failures in service.

SLIDE 13 Crystal Defects — Volume Defects

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Grain boundaries are high-energy regions that actively control strength, toughness, corrosion, and diffusion — not just where grains meet.

$$\sigma_y = \sigma_0 + k / \sqrt{d}$$

- When a dislocation reaches a grain boundary, it cannot cross — orientation mismatch blocks it
- Dislocations pile up, creating a back-stress that resists further deformation
- Smaller grain = shorter distance before hitting a boundary = more pile-ups = stronger
- One of the most powerful and practical strengthening tools in metallurgy

1. Recovery

2. Recrystallization

3. Grain Growth

New grains coarsen as boundaries migrate to reduce total energy. Control time and temperature to reach target grain size.

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Even in solid metal, atoms are not perfectly still — they vibrate constantly, and at elevated temperatures they have enough energy to jump between sites.

Larger atoms move by jumping into neighboring vacancies — hopping from one lattice site to an adjacent empty one.

Rate: Slow — vacancies are relatively rare and large atoms must squeeze between others.

Small atoms move through the gaps between host atoms without displacing them — flowing through the interstitial network.

Rate: Fast — gaps are abundant, and small atoms can move easily even at lower temperatures.

Example: Carbon in iron (carburizing, case hardening),
nitrogen in steel (nitriding)

Diffusion rate increases exponentially with temperature — a small temperature rise causes a dramatic increase in atom movement. This is why heat treatment temperatures must be controlled precisely.

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Core Questions — Answers

- 1 What is the difference between strength and toughness?
Strength is the peak stress before yielding or fracture. Toughness is total energy absorbed — the area under the stress-strain curve. A material can be strong but brittle (high-carbon steel, glass): high peak stress, negligible plastic zone, low toughness.
- 2 What happens at the yield point at the atomic level?
Dislocations overcome their pinning obstacles and begin moving irreversibly through the crystal. Below yield, bonds stretch elastically and recover. At yield, atomic planes shift permanently — plastic deformation begins and cannot be undone by removing the load.
- 3 Why does cold work increase strength AND reduce ductility simultaneously?
Cold work multiplies dislocation density enormously. Dislocations tangle and block each other — work hardening — requiring more stress to continue deformation (higher strength). But with so many dislocations already present, little capacity for further plastic flow remains (lower ductility).
- 4 Fine-grained vs. coarse-grained steel: which is stronger and why?
Fine-grained. Hall-Petch: $\sigma_y = \sigma_0 + k/\sqrt{d}$. More grain boundaries per unit volume block dislocation movement at each boundary — each crossing requires additional stress. Finer grains = more barriers = higher yield strength.
- 5 Shaft fails below yield strength under cyclic loading. What is it?
Fatigue failure. Cyclic loading initiates tiny cracks at stress concentrations (surface defects, notches) that grow with each cycle until catastrophic fracture — often well below the static yield strength. The relevant property is the endurance limit.

SLIDE 16 Core Questions — Answers

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Key Takeaways

1 Stress and strain are the engineer's language

Stress ($\sigma = F/A$) is internal force per area. Strain ($\epsilon = \Delta L/L_0$) is deformation relative to original size. The yield point — the elastic-to-plastic transition — is the most important value in structural material selection.

2 Defects are features, not failures

Point, line, planar, and volume defects each affect metal differently. Dislocations enable ductility — eliminate them and the metal becomes brittle. Real engineering metals are defined by their defect structure.

3 Grain size is the most practical lever

Hall-Petch: $\sigma_y = \sigma_0 + k/\sqrt{d}$ — making grains smaller makes metal stronger. Grain size is controlled through alloy additions and thermal processing (annealing temperature and time).

4 Diffusion drives heat treatment

Atoms move through solids at rates exponential with temperature. Carburizing, nitriding, and homogenization all rely on controlled diffusion — this is why heat treatment temperatures are precise.

5 Safety is not optional!

PPE is mandatory in the shop — no exceptions. We work with heat, sharp edges, and hazardous materials every lab session. Protect yourself and your classmates.

SLIDE 17 Key Takeaways

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